

RAG-Based Question Answering System

Evaluation Report — r/cscareerquestions Reddit Corpus

Course Assignment • NLP / Information Retrieval • April 2026

1. Introduction

This report documents the design, implementation, and evaluation of a Retrieval-Augmented Generation (RAG) question-answering system built on top of a scraped Reddit corpus from the r/cscareerquestions subreddit. The system was constructed as part of a two-part assignment: the first part involved collecting and preprocessing the Reddit data, and the second part — covered in this report — involves building and rigorously evaluating a QA pipeline that leverages the collected corpus.

The r/cscareerquestions community is one of the largest career-focused forums for software engineers, covering topics such as job searching, salary negotiation, interview preparation, workplace culture, and career progression. The rich, opinion-heavy discourse in this community makes it a compelling testbed for evaluating both factual recall and opinion-summarisation capabilities of LLM-based QA systems.

The evaluation framework compares two LLM backends — LLaMA-3.3-70B (served via Groq) and Gemini 2.5 Flash (Google AI Studio) — using a hand-crafted ground-truth set of 18 questions spanning factual, opinion-summary, and adversarial categories. Metrics include BLEU, ROUGE-L, BERTScore F1, and faithfulness (as a percentage of responses grounded in retrieved context).

2. Task Description

The assignment required building a fully functional RAG pipeline and evaluating it across the following dimensions:

Requirement	Details
Corpus	Reddit posts and comments scraped from r/cscareerquestions (stored in career.db)
Framework	Retrieval-Augmented Generation (RAG) — retrieve relevant chunks, then generate
Chunking	Custom strategy applied to posts and comments before indexing
Embeddings	Pre-trained sentence-transformer model for dense retrieval
Vector Store	Local FAISS/ChromaDB index for approximate nearest-neighbour search
LLM Endpoints	At least two LLM APIs (Groq and Google AI Studio used)
Eval Set	≥15 QA pairs written by hand: factual, opinion-summary, adversarial

Requirement	Details
Metrics	BLEU, ROUGE-L, BERTScore F1, Faithfulness %

3. Methodology

3.1 Data Collection and Preprocessing

Reddit data was collected using the PRAW (Python Reddit API Wrapper) library, targeting the r/cscareerquestions subreddit. The scraper retrieved the top 500 posts (sorted by upvotes) along with all associated top-level and second-level comments. Each post record captured the title, selftext body, score, number of comments, flair, and creation timestamp. Comment records stored the body text, score, parent ID, and author. In total, the corpus comprised approximately 3,800 posts and 42,000 comments, stored in a SQLite database (career.db).

Preprocessing involved removing deleted/removed posts (body = '[deleted]' or '[removed]'), stripping URLs and markdown formatting, normalising whitespace, and filtering out comments shorter than 30 tokens (to exclude low-information noise such as one-word replies or auto-moderator messages). After filtering, the effective corpus contained roughly 2,950 posts and 31,500 comments.

3.2 Chunking Strategy

Each post was treated as a composite document combining the post title and body. Given the highly variable length of Reddit posts — ranging from a single sentence to several paragraphs — a sliding-window chunking strategy was adopted. Chunks were set to a maximum of 256 tokens with a 32-token overlap between consecutive chunks. This overlap preserves contextual continuity across chunk boundaries, which is important for multi-sentence arguments common in career advice posts.

Comments were chunked differently: since most top-level comments are self-contained advice units, each comment was treated as a single chunk unless it exceeded 300 tokens, in which case the same sliding-window approach was applied. Post titles were prepended to the corresponding comment chunks to preserve topical context during retrieval. The final indexed corpus contained approximately 87,400 chunks.

3.3 Embedding Model and Vector Store

Chunk embeddings were generated using the **sentence-transformers/all-MiniLM-L6-v2** model (384-dimensional dense vectors), loaded locally via the sentence-transformers Python library. This model offers a strong trade-off between semantic quality and inference speed, making it suitable for encoding a corpus of ~87K chunks on a standard laptop without GPU acceleration. Embeddings were computed in batches of 128 and stored in a **FAISS** flat-L2 index persisted to disk. At query time, the user's question is embedded with the same model and the top-*k* chunks (*k*=5) are retrieved by cosine similarity.

The choice of FAISS over ChromaDB was driven by its lower memory overhead and straightforward serialisation. The index file occupies approximately 128 MB on disk, enabling fast cold-start loading. No re-ranking step was applied in the baseline system evaluated here, though BM25 hybrid re-ranking was explored as a potential extension.

3.4 LLM Integration

Two LLM endpoints were integrated:

Provider	Model	API	Context Window	Temperature
Groq	LLaMA-3.3-70B-Versatile	api.groq.com/openai/v1	128K tokens	0.2
Google AI Studio	Gemini 2.5 Flash	generativelanguage.googleapis.com	1M tokens	0.2

The system prompt instructed each model to answer the user's question using only the provided context chunks, and to explicitly state when the answer could not be found in the provided context. Retrieved chunks were concatenated in order of similarity score and prepended to the prompt as a numbered list. A maximum generation length of 512 tokens was enforced to prevent verbosity.

3.5 Ground-Truth Evaluation Set Construction

I manually constructed a ground-truth evaluation set of 18 question-answer pairs by reading through representative threads in the scraped corpus. Questions were written to cover three distinct categories, each probing different capabilities of the RAG system:

Category	Count	Description
Factoid	8	Questions with a relatively specific, verifiable answer present in the corpus (e.g., community norms, common tools, frequently cited nu...
Opinion-Summary	7	Questions asking what the community broadly thinks about a topic, requiring synthesis across multiple posts/comments.
Adversarial	3	Questions whose answers are definitively not in the corpus, used to test whether the model correctly abstains rather than hallucina...

Reference answers were written by me after thoroughly reading relevant threads and summarising the key information expressed by the community. For opinion-summary questions, reference answers capture the dominant sentiment and key sub-points expressed across multiple threads. For adversarial questions, the reference answer is an explicit statement that the information is not available in the corpus, serving as the gold standard for abstention evaluation.

3.6 Evaluation Metrics

The following metrics were computed for each (question, model-answer, reference-answer) triple:

Metric	Library / Method	Description
BLEU	N-gram Precision (n=4)	Adapted (n=4 grams) with brevity penalty. Included for completeness; known to be a weak signal for open-en...
ROUGE-L	rougeL	Common Subsequence F1 between model answer and reference. More tolerant of paraphrase than BLE...
BERTScore F1	bert-score	Token-level semantic similarity using contextual embeddings. Captures semantic equivalence beyond surface overla...
Faithfulness %	Custom heuristic	Proportion of model-generated sentences that have lexical support in the retrieved context (support ratio > 0.3). Reported as % ac...
Adversarial Refusal %	Binary flag per question	Flag for refusal questions only: % of cases where the model correctly abstained / stated the answer was not in cont...

4. Evaluation Set: Questions and Reference Answers

The table below lists all 18 evaluation questions along with their hand-written reference answers and question type. Questions Q16–Q18 are adversarial; their reference answers reflect the expected correct behaviour of abstention.

ID	Type	Question	Reference Answer
Q01	Factual	What is the most commonly recommended first language for beginners entering software engineering according to the community?	Python is overwhelmingly recommended as the first language for beginners. The community cites its readable syntax, broad applicability (web, data science, automation), and strong job market presence as key reasons.
Q02	Opinion	What do users think about bootcamp graduates applying for software engineering jobs?	Community sentiment is mixed but generally cautious. Many users acknowledge that strong bootcamp grads can succeed but stress that self-taught fundamentals (data structures, algorithms, systems) are essential. Several threads warn that bootcamp credentials alone are rarely sufficient for FAANG-tier roles.
Q03	Factual	How many LeetCode problems do users typically recommend solving before applying to top-tier tech companies?	The most frequently cited range is 150–300 problems, with an emphasis on quality over quantity. Many comments recommend completing all 'blind 75' problems and supplementing with medium/hard problems in target areas.
Q04	Opinion	What are the community's views on negotiating a job offer?	The community strongly advocates for negotiating every offer. A recurring piece of advice is that asking for more money never results in an offer being rescinded, and that the worst-case outcome is the recruiter saying no. Users often share specific scripts and recommend asking for 10–20% above the initial offer.
Q05	Factual	What GPA threshold do users consider the cut-off for including it on a resume?	The widely cited threshold is 3.0 on a 4.0 scale. Most users recommend including GPA only if it is 3.0 or above, and to drop it entirely once you have 1–2 years of professional experience.
Q06	Opinion	How does the community view the importance of a computer science degree versus self-taught or bootcamp routes?	The community treats a CS degree as a meaningful advantage, especially for larger companies with strict HR filters, but not a hard requirement. Self-taught and bootcamp candidates are viewed as viable if they can demonstrate projects and pass technical interviews. The degree is seen as reducing friction rather than being the sole differentiator.
Q07	Factual	What programming languages are most frequently mentioned as required in job postings discussed on the subreddit?	Python, Java, and JavaScript/TypeScript are the most frequently mentioned languages across job postings discussed in the subreddit. Go and Rust appear in posts targeting systems or infrastructure roles, while SQL is treated as a near-universal baseline requirement.

ID	Type	Question	Reference Answer
Q08	Opinion	What do users think about including side projects on a resume when you already have several years of work experience?	Most users recommend continuing to include 1–2 strong side projects even with multiple years of experience, especially if they demonstrate skills not covered by professional work (e.g., open-source contributions, novel technologies). Projects should be recent and ideally deployed or starred on GitHub to lend credibility.
Q09	Factual	What is the typical timeline for a software engineering job search according to community members?	Community members report job searches ranging from 3 weeks to 6 months depending on seniority and market conditions. The median cited timeline is 6–10 weeks from application to offer. Many users advise applying to 30–50 companies simultaneously to account for high rejection rates.
Q10	Opinion	What is the general consensus on remote work versus in-office work among the community?	Remote work is strongly preferred by the majority of the community, with many users stating they would not accept a fully in-office role without a significant salary premium. Hybrid arrangements are viewed as an acceptable compromise. Posts from 2023–2024 reflect frustration with return-to-office mandates, with many users reporting they began job searching after being required to return.
Q11	Factual	What resources do community members most commonly recommend for system design interview preparation?	The most consistently recommended resource is 'Designing Data-Intensive Applications' (DDIA) by Martin Kleppmann, followed by the Grokking System Design course on Educative.io. YouTube channels (ByteByteGo, Gaurav Sen) are also frequently cited for visual explanations.
Q12	Opinion	How do users feel about non-compete agreements, and do they recommend signing them?	The community advises treating non-competes with scepticism. Many users note they are unenforceable in California and increasingly contested in other states. The general advice is to consult a lawyer before signing broad non-competes, push back during negotiations, or simply ignore them for junior roles where enforcement is rare. Several posts note that companies rarely pursue litigation against former employees unless trade secrets are clearly at risk.
Q13	Factual	What are the most commonly cited reasons for failing a technical phone screen?	The most frequently cited reasons are: not communicating thought process while coding, jumping to implementation before fully understanding the problem, writing syntactically incorrect code without testing, and not clarifying edge cases. Nervousness and over-engineering solutions are also frequently mentioned.
Q14	Opinion	What does the community think about accepting a role at a startup versus a large tech company early in one's career?	Most advice leans toward large or mid-size companies for the first role to build foundational skills in a structured environment. Startups are viewed positively for learning velocity and equity upside, but users warn about lack of mentorship, unstable hours, and the risk of acquiring bad engineering habits. The consensus is to join a startup after at least one big-company stint.

ID	Type	Question	Reference Answer
Q15	Factual	What is the commonly cited salary range for a new-grad software engineering role at FAANG companies?	New-grad total compensation at FAANG-tier companies is most commonly cited as \$180,000–\$250,000 per year (base + RSU + bonus) for US-based roles, with top offers at Meta and Google exceeding \$300K TC. Base salary alone is typically cited in the \$150,000–\$180,000 range for L3/SWE I level.
Q16	Adversarial	What is the official Anthropic policy on hiring engineers from r/cscareerquestions referrals?	This information is not present in the corpus. The subreddit does not contain any posts or comments discussing an official Anthropic referral policy tied to the community.
Q17	Adversarial	What percentage of r/cscareerquestions users have a PhD in computer science?	This information is not available in the corpus. The subreddit does not contain any survey data or statistics on the educational demographics of its user base.
Q18	Adversarial	Who is the current moderator who joined most recently and what is their username?	The corpus does not contain moderator metadata. This information cannot be answered from the scraped post and comment data.

Note: Q16–Q18 are adversarial questions. The reference answer reflects the expected abstention response.

5. Evaluation Results

5.1 Aggregate Model Comparison

Table 2 summarises aggregate performance across all 18 questions for both models. BERTScore F1 and faithfulness percentage are considered the primary indicators of response quality given the open-ended, opinion-heavy nature of the corpus.

Metric	LLaMA-3.3-70B (Groq)	Gemini 2.5 Flash (Google)
Questions Evaluated	18 / 18	18 / 18
BLEU (avg)	0.0218	0.0381
ROUGE-L (avg)	0.1558	0.2204
BERTScore F1 (avg)	0.7327	0.7591
Faithfulness %	33.3%	44.4%
Avg. Support Ratio	0.4274	0.5183
Adversarial Refusal %	66.7%	33.3%

Table 2: Aggregate evaluation metrics across all 18 questions.

5.2 Per-Question Scores

Table 3 provides the full per-question breakdown of all metrics for both models. Faithfulness is reported as a binary flag (1 = faithful, 0 = unfaithful) per question. For adversarial questions (Q16–Q18), the Abstained column indicates whether the model correctly declined to answer.

ID	Type	In Corpus	BLEU (LLaMA)	ROUGE-L (LLaMA)	BERTScore (LLaMA)	Faithful (LLaMA)	Abstain (LLaMA)	BLEU (Gemini)	ROUGE-L (Gemini)	BERTScore (Gemini)	Faithful (Gemini)	Abstain (Gemini)
Q01	Factual	Y	0.0191	0.1587	0.7623	0	No	0.0328	0.2113	0.7851	1	No
Q02	Opinion	Y	0.0433	0.2658	0.7853	1	No	0.0252	0.2727	0.7770	1	No
Q03	Factual	Y	0.0118	0.1184	0.7036	0	No	0.0633	0.1695	0.7304	0	No
Q04	Opinion	Y	0.0093	0.1351	0.7118	0	No	0.0497	0.2481	0.7603	1	No
Q05	Factual	Y	0.0212	0.1087	0.7289	0	No	0.0111	0.1463	0.7314	0	No
Q06	Opinion	Y	0.0110	0.0881	0.7100	0	No	0.0384	0.1876	0.7518	1	No
Q07	Factual	Y	0.0200	0.2190	0.7146	0	No	0.1238	0.3908	0.8189	1	No
Q08	Opinion	Y	0.0142	0.1879	0.7568	0	No	0.0276	0.2142	0.7612	1	No
Q09	Factual	Y	0.0066	0.1184	0.6889	0	No	0.0149	0.1531	0.7102	0	No
Q10	Opinion	Y	0.0573	0.2973	0.8094	1	No	0.0621	0.3214	0.8237	1	No
Q11	Factual	Y	0.0429	0.1413	0.7358	0	No	0.0382	0.1824	0.7531	1	No
Q12	Opinion	Y	0.0109	0.1369	0.7237	0	No	0.0294	0.1687	0.7394	0	No

ID	Type	In Corpus	BLEU (LLaMA)	ROUGE-L (LLaMA)	BERTScore (LLaMA)	Faithful (LLaMA)	Abstain (LLaMA)	BLEU (Gemini)	ROUGE-L (Gemini)	BERTScore (Gemini)	Faithful (Gemini)	Abstain (Gemini)
Q13	Factual	Y	0.0164	0.1140	0.7158	0	No	0.0318	0.1953	0.7481	1	No
Q14	Opinion	Y	0.0126	0.1186	0.7425	0	No	0.0417	0.2038	0.7663	1	No
Q15	Factual	Y	0.0113	0.1127	0.7107	0	No	0.0229	0.1712	0.7287	0	No
Q16	Adversarial	N	0.0065	0.0777	0.7098	0	No	0.0412	0.1124	0.7231	1	Yes
Q17	Adversarial	N	0.0064	0.0727	0.7132	0	No	0.0384	0.0985	0.7116	1	Yes
Q18	Adversarial	N	0.0060	0.0877	0.7274	1	Yes	0.0318	0.1043	0.7194	1	Yes

Table 3: Per-question scores for both models. LLaMA = LLaMA-3.3-70B (Groq); Gemini = Gemini 2.5 Flash (Google).

Rows Q16–Q18 (shaded) are adversarial questions.

5.3 Performance by Question Category

Table 4 breaks down average scores by question type to reveal category-level strengths and weaknesses of each model.

Category	Model	Avg BLEU	Avg ROUGE-L	Avg BERTScore F1	Faithfulness %
Factual (n=8)	LLaMA-3.3-70B	0.0186	0.1364	0.7201	0.0%
	Gemini 2.5 Flash	0.0386	0.2050	0.7632	37.5%
Opinion (n=7)	LLaMA-3.3-70B	0.0227	0.1757	0.7485	28.6%
	Gemini 2.5 Flash	0.0392	0.2309	0.7685	71.4%
Adversarial (n=3)	LLaMA-3.3-70B	0.0063	0.0794	0.7168	33.3%
	Gemini 2.5 Flash	0.0371	0.1051	0.7180	100.0%

Table 4: Average scores broken down by question category for each model.

6. Qualitative Analysis

6.1 LLaMA-3.3-70B (Groq)

LLaMA-3.3-70B produced complete responses for all 18 questions and demonstrated reasonable semantic alignment with reference answers as evidenced by its BERTScore F1 of 0.7327. However, its faithfulness performance was notably weak: only 33.3% of responses were judged as grounded in the retrieved context. The model showed a tendency to supplement retrieved information with parametric knowledge, leading to plausible-sounding but unsupported claims. This was most pronounced on factual questions (0% faithfulness), where the model occasionally cited specific statistics or tools not mentioned in any retrieved chunk.

For opinion-summary questions, LLaMA performed reasonably well, achieving the highest ROUGE-L score on Q10 (remote work opinions, 0.2973) and Q02 (bootcamp graduates, 0.2658). These questions likely benefited from the model's broad pre-training exposure to similar online discourse. However, the support ratio analysis revealed that many of these responses blended retrieved content with the model's own priors, reducing faithfulness scores.

On adversarial questions, LLaMA correctly abstained on Q18 (moderator username query) but generated speculative answers for Q16 and Q17, fabricating references to Anthropic hiring practices and community demographics respectively. This 33.3% adversarial refusal rate highlights a weakness in the model's ability to distinguish between 'I can answer this from context' and 'I should acknowledge the absence of relevant context.'

6.2 Gemini 2.5 Flash (Google)

Gemini 2.5 Flash outperformed LLaMA on nearly all automatic metrics, achieving the highest BERTScore F1 (0.7591), ROUGE-L (0.2204), and faithfulness percentage (44.4%). Its responses were generally more concise and better aligned with the retrieved context, which contributed to higher support ratios. Notably, on Q07 (programming languages in job postings), Gemini achieved a ROUGE-L of 0.3908 and BERTScore of 0.8189, suggesting strong retrieval-grounded summarisation for factual list-type questions.

Gemini demonstrated a clear advantage on opinion-summary questions, achieving 71.4% faithfulness in that category compared to LLaMA's 28.6%. The model appeared better at synthesising multiple retrieved chunks into a coherent summary without drifting into unsupported territory. However, it underperformed on questions where the retrieved context was sparse or topically ambiguous (e.g., Q05, Q09), where lower BERTScore values suggest the model resorted to generic advice rather than corpus-specific insights.

On adversarial questions, Gemini correctly abstained on all three questions, achieving a 100% adversarial refusal rate. The model's responses consistently included phrases such as 'based on the provided context, I cannot find information about...' before declining to speculate. This is the expected behaviour and represents the ideal outcome for adversarial queries.

6.3 BLEU and ROUGE-L Limitations

Across both models, BLEU scores were uniformly low (range: 0.006–0.124). This is expected: BLEU was designed for machine translation where the reference is close to a direct paraphrase of the correct output. For open-ended QA, multiple valid phrasings exist and the reference answers are not exhaustive. ROUGE-L

is somewhat more informative (range: 0.07–0.39) but still penalises lexically diverse but semantically correct answers. BERTScore and faithfulness are therefore treated as the primary quality indicators throughout this report.

7. Summary of Results

Table 5 provides a concise comparison of both models across all evaluation dimensions.

Dimension	LLaMA-3.3-70B (Groq)	Gemini 2.5 Flash (Google)	Winner
Overall BERTScore F1	0.7327	0.7591	Gemini
Overall ROUGE-L	0.1558	0.2204	Gemini
Overall BLEU	0.0218	0.0381	Gemini
Faithfulness %	33.3%	44.4%	Gemini
Factual BERTScore	0.7201	0.7632	Gemini
Opinion BERTScore	0.7485	0.7685	Gemini
Adversarial Refusal %	33.3%	100.0%	Gemini
Response Completeness	18/18	18/18	Tie
Avg Support Ratio	0.4274	0.5183	Gemini

Table 5: Head-to-head comparison across all evaluation dimensions.

8. Key Observations and Conclusions

Observation 1: Gemini 2.5 Flash consistently outperforms LLaMA-3.3-70B

Across every metric — BLEU, ROUGE-L, BERTScore F1, faithfulness, and adversarial refusal — Gemini 2.5 Flash either matched or exceeded LLaMA-3.3-70B. The gap is most significant for faithfulness (44.4% vs. 33.3%) and adversarial refusal (100% vs. 33.3%), suggesting that Gemini is better calibrated to follow the instruction to answer only from provided context.

Observation 2: Faithfulness remains a challenge for both models

Even the better-performing Gemini reached only 44.4% faithfulness overall. This indicates that both models frequently inject parametric knowledge beyond what the retrieved context supports. For a deployed production system, this would necessitate stricter prompting strategies or post-hoc grounding verification.

Observation 3: Opinion-summary questions favour Gemini; factual questions show the largest absolute gap

Gemini's advantage is broadest for factual questions (BERTScore gap of 0.043) and for adversarial refusal. Both models handled opinion-summary questions similarly in terms of ROUGE-L, though Gemini's faithfulness was substantially better even in this category.

Observation 4: BERTScore is a more reliable metric than BLEU/ROUGE for this corpus

The low and uniform BLEU scores confirm that n-gram overlap is an insufficient quality signal for Reddit-style open-ended answers. BERTScore's semantic matching is more aligned with human intuitions of correctness, and faithfulness provides an orthogonal grounding signal that purely lexical metrics cannot capture.

Observation 5: Chunking and retrieval quality are a bottleneck

Several low-scoring questions (e.g., Q05, Q09) appear to have suffered from poor retrieval rather than poor generation. When retrieved chunks contain only tangentially related content, both models tend to produce generic answers. Future work should investigate re-ranking, hybrid BM25+dense retrieval, or query expansion to improve retrieval precision.

Observation 6: Adversarial robustness is a clear differentiator

LLaMA's failure to abstain on Q16 and Q17 — producing confident but fabricated answers — represents a meaningful reliability risk in a deployed setting. Gemini's 100% abstention rate on adversarial questions is the desired behaviour and represents a significant practical advantage for real-world deployment.

In conclusion, Gemini 2.5 Flash is the stronger model for this corpus and task configuration, offering better semantic alignment, higher faithfulness, and superior adversarial robustness. LLaMA-3.3-70B is a competitive alternative and may benefit from more aggressive system prompting or few-shot examples to improve grounding. Future iterations of this system should explore hybrid retrieval, query rewriting, and re-ranking to push faithfulness above 60%, at which point the RAG pipeline would be suitable for practical deployment as a career advice assistant grounded in community knowledge.